بسم الله الرحمن الرحيم



Al Balqa Applied University
Faculty of engineering Technology
Mechatronics Engineering Department
Lecture notes in Power semiconductor drive of AC Motors

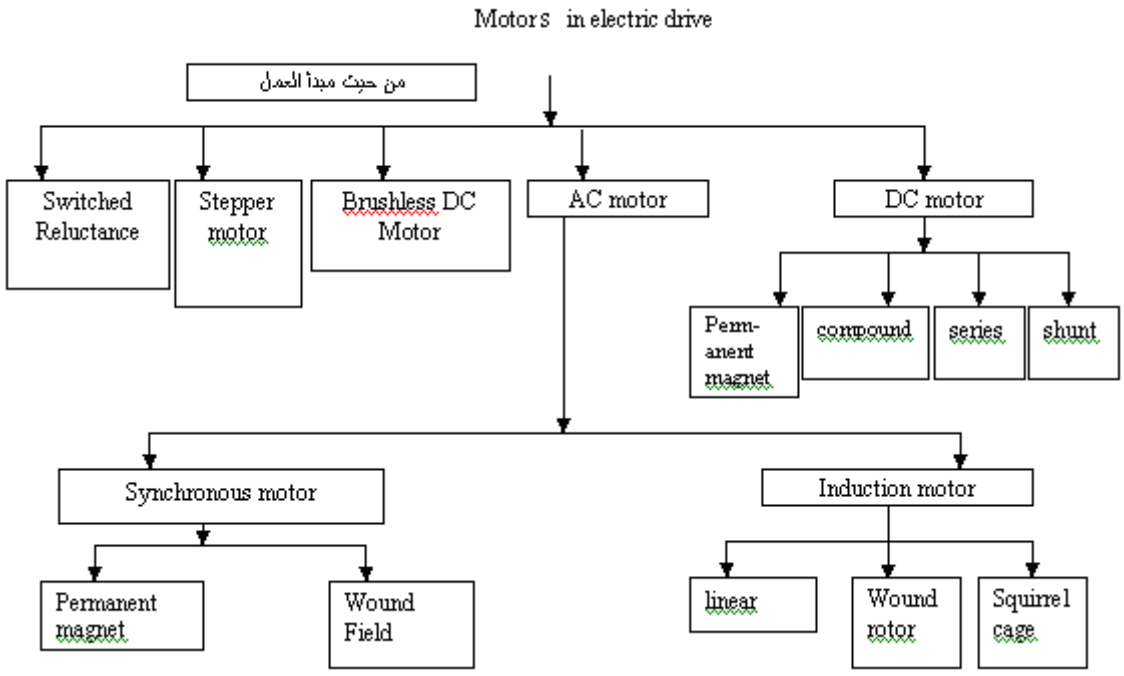
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Introduction

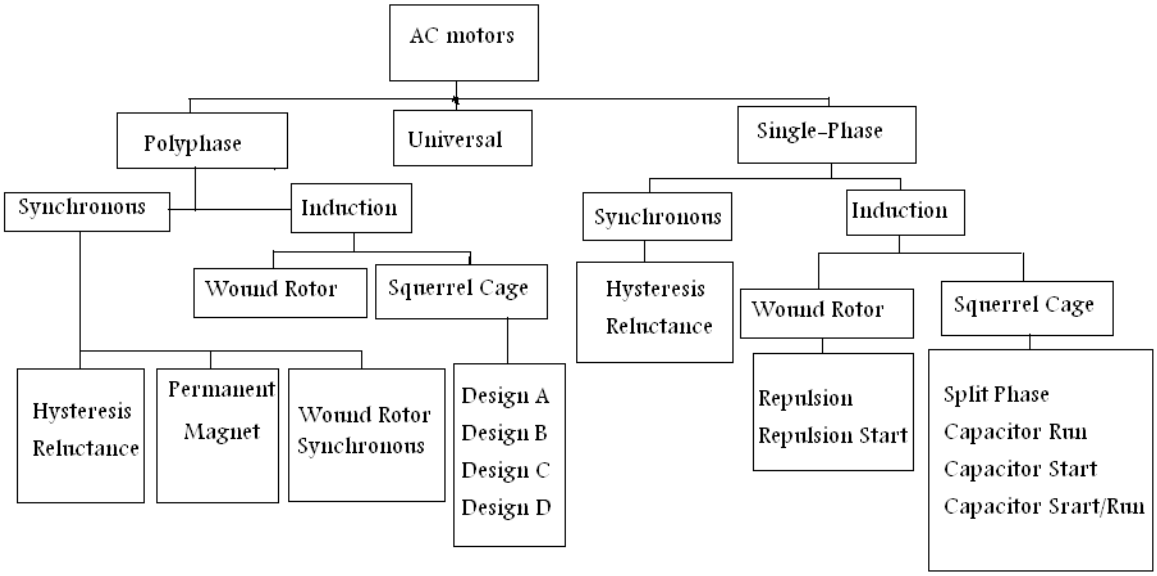
An important factor in industrial progress during the past five decades has been the increasing sophistication of factory automation which has improved productivity manifold. Manufacturing lines typically involve a variety of variable speed motor drives which serve to power conveyor belts, robot arms, overhead cranes, steel process lines, paper mills, and plastic and fiber processing lines to name only a few. Prior to the 1950s all such applications required the use of a DC motor drive since AC motors were not capable of smoothly varying speed since they inherently operated synchronously or nearly synchronously with the frequency of electrical input. To a large extent, these applications are now serviced by what can be called general-purpose AC drives. In general, such AC drives often feature a cost advantage over their DC counterparts and, in addition, offer lower maintenance, smaller motor size, and improved reliability. However, the control flexibility available with these drives is limited and their application is, in the main, restricted to fan, pump, and compressor types of applications where the speed need be regulated only roughly and where transient response and low-speed performance are not critical. More demanding drives used in machine tools, spindles, high-speed elevators, dynamometers, mine winders, rolling mills, glass float lines, and the like have much more sophisticated requirements and must afford the flexibility to allow for regulation of a number of variables, such as speed, position, acceleration, and torque. Such high-performance applications typically require a high speed holding accuracy better than 0.25%, a wide speed range of at least 20:1, and fast transient response, typically better than 50 rad/s, for the speed loop. Until recently, such drives were almost exclusively the domain of DC motors combined with various configurations of AC-to-DC converters depending upon the application. With suitable control, however, induction motor drives 2 AC Motor Speed Control Draft Date: February 5, 2002 have been shown to be more than a match for DC drives in high-performance applications. While control of the induction machine is considerably more complicated than its DC motor counterpart, with continual advancement of microelectronics, these control complexities have essentially been overcome. Although induction motors drives have already overtaken DC drives during the next decade it is still too early to determine if DC drives will eventually be relegated to the history book. However, the future decade will surely witness a continued increase in the use of AC motor drives for all variable speed applications.

1.1Motors' classification:

The following tree shows the classification of electric motors:



Classification of AC motors:
According to the supply voltage, AC motors can be classified as in the following tree:



Induction motors

The most widely used machine in industry is the three-phase induction motor. Its characteristic features are:

- Simple and rugged construction
- Low cost and minimum maintenance
- High reliability and sufficiently high efficiency
- Needs no extra starting motor and need not be synchronized

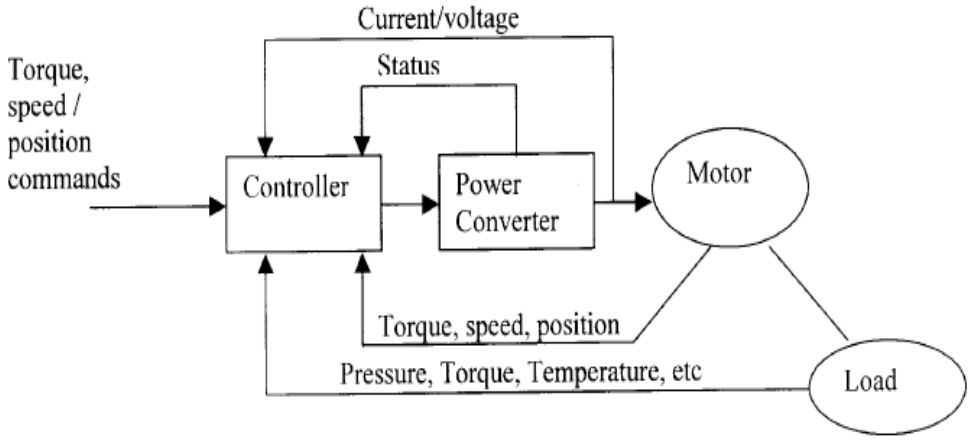
An Induction motor has basically two parts – Stator and Rotor

The Stator is made up of a number of stampings with slots to carry three phase windings. It is wound for a definite number of poles. The windings are geometrically spaced 120 degrees apart. Two types of rotors are mostly used in Induction motors-

Squirrel-cage rotor and Wound rotor

A squirrel-cage rotor consists of thick conducting bars embedded in parallel slots. These bars are short-circuited at both ends by means of short-circuiting rings. A wound rotor has three-phase, double-layer, distributed winding. It is wound for as many poles as the stator. The three phases are wired internally and the other ends are connected to slip-rings mounted on shaft with brushes resting on them. The brushes are connected to an external resistance that does not rotate with the rotor and can be varied to change the N-T characteristics. In fact, an Induction motor can be compared with a transformer because of the fact that just like a transformer it is a singly energized device which involves changing flux linkages with respect to a primary (stator) winding and secondary(rotor) winding.

Motor Drive Scheme:

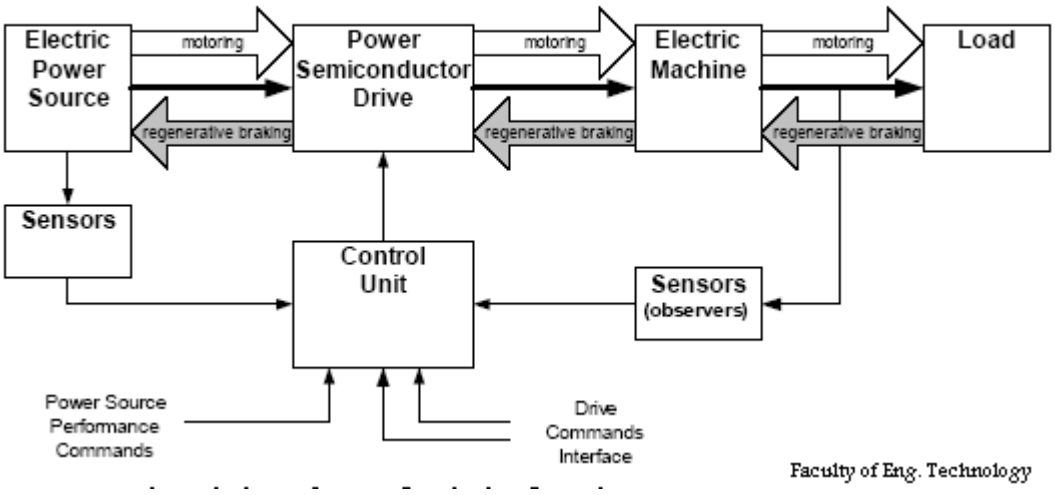


Basic ASD Components

Most adjustable-speed drive (ASD) systems consist of three basic parts. A rectifier that converts the fixed frequency ac input voltage to dc. An inverter that switches the rectified dc voltage to an adjustable frequency ac output voltage (The inverter may also control output current flow, if desired). The dc link connects the rectifier to the inverter. A set of controls directs the rectifier and inverter to produce the desired ac frequency and voltage to meet the needs of the ASD system at any moment in time.

Modern electric drive:

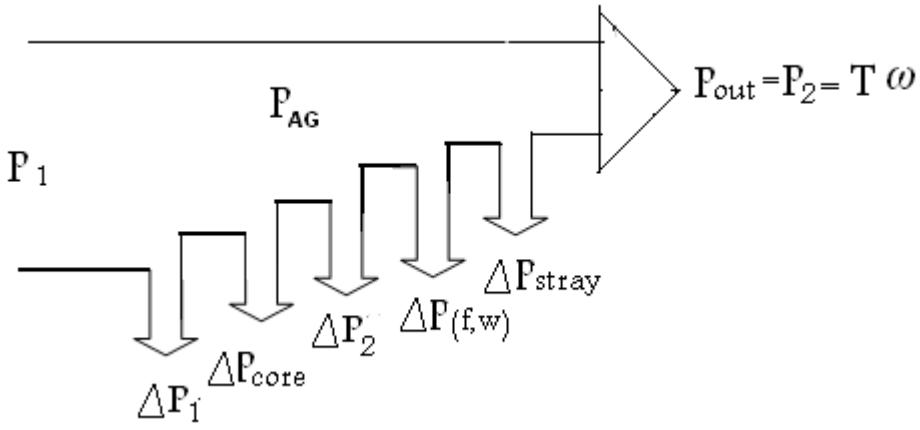
Modern electric drive uses power semiconductor drive which includes semiconductor elements such as thyristors, diodes, transistors and so on. Nowadays, IGBTs (isolated gate bipolar transistors) are used widely in electric drive because of their good merits comparing with thyristors. The figure below shows the modern electric drive system.



Losses and the power flow diagram

$P_1 = \sqrt{3}.V_l I_l \cos\theta$

$\Delta P_1 = \Delta P_{scl} = 3I_1^{\,2} R_1$



$\Delta P_1 = \Delta P_{scl}$ = stator copper losses.

ΔP_{1core} = *losses due to hysteretic and eddy current in the stator.*

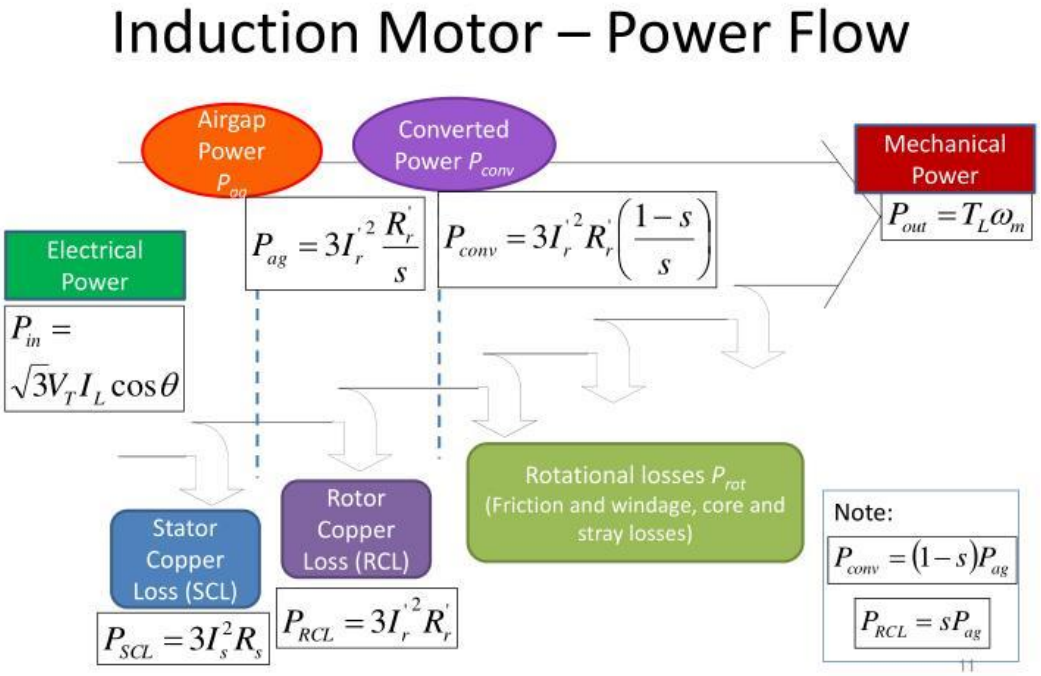
P_{AG} = *air gap power.*

$\Delta P_2 = \Delta p_{rcl}$ = *rotor copper losses;* $P_{rcl} = 3\,(I_2)^2\,R_2$

P_{con} = *power converted from electrical to mechanical form.*

$\Delta P_{(f,w)}$ = *friction and windage losses,* ΔP_{stray} = *stray losses.*

$\Delta P_{core} = \Delta P_{1\,core} + \Delta P_{2\,core} \approx \Delta P_{1core}$



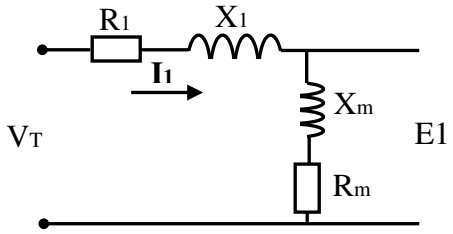
Because of the relation motion of the magnetic field over the rotor surface is quite slow, the rotor core losses are very tiny compared to the stator core losses.

The higher the speed of an induction motor, the higher is friction, windage and stray losses, on the other hand, the higher the speed of the motor, the lower is core losses.
Therefore, these three types of losses are some times called rotational losses.
Rotational losses of a motor are often considered to be constant with changing speed, since the component losses change in opposite direction with changing in speed.

الدائرة المكافئة للطور الواحد في المحرك
Equivalent circuits for the induction motor

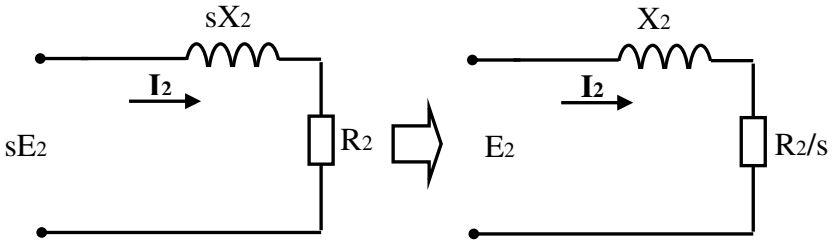
Stator circuit:-

R_1 – per phase stator winding resistance.
X_1 - per phase stator leakage resistance.
E_1 – per phase induced voltage in the stator winding.
R_m – per phase magnetizing resistance.
X_m – per phase magnetizing leakage



Rotor circuit:

The following circuits represent the equivalent rotor circuit where:
R_2 – per phase rotor resistance.
X_2 - per phase rotor leakage resistance.
E_2 – per phase induced voltage in the rotor circuit at standstill condition (blocked rotor).
s – motor slip

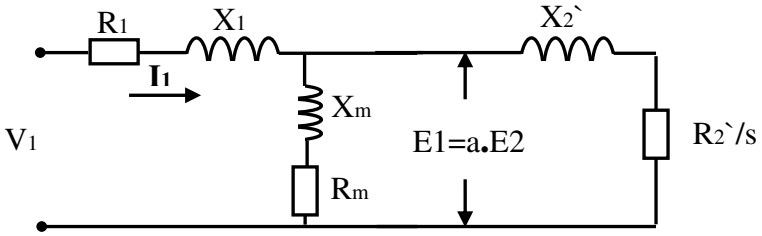


$$s = \frac{n_0 - n}{n_0} = \frac{\omega_0 - \omega}{\omega_0},\ \omega = \frac{2\pi n}{60},\ n_0 = \frac{60 f_1}{p}$$

Equivalent circuit type T:

The following circuit can be obtained by combining stator and rotor circuits where:.

R'$_2$ - referred per phase rotor circuit resistance.
X'$_2$ - referred per phase rotor circuit reactance.
a- turns' ratio



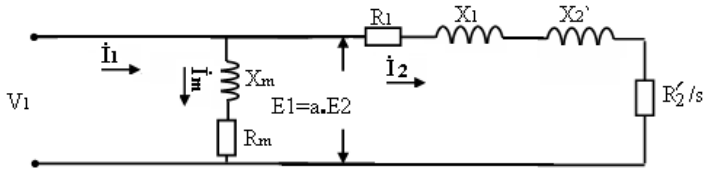
T- Type

$$R_2' = a^2 R_2, X_2' = a^2 X_2, I_2' = \frac{I_2}{a}$$

$$\boldsymbol{a = \frac{N_1}{N_2} = \frac{E_1}{E_2}}$$

Equivalent circuit Type Π:

Eq. circuit type T can be replaced by eq. circuit type π which considered more convenient for motor parameters' calculations,



π - type : per-phase stator referred equivalent circuit of an induction motor

where:

$R_1 = k.R_{1T}$

$X_1 = k.X_{1T}$

$R'_2 = k^2.R'_{2T}$

$X'_2 = k^2.X'_{2T}$

$Rm = Rm_T + R_{1T}$

$Xm = Xm_T + X_{1T}$

$k=1+(X_{1T}/Xm_T)-j(R_{1T}/Xm_T)$, While R_{1T} is very small comparing with X_{mT} so:

$k=1+(X_{1T}/Xm_T)=(1.05\text{-}1.1)$

$$\frac{R_2}{S} = R_2 + \frac{R_2(1-S)}{S}$$

حيث: $\mathbf{R_2}$ تمثل مقاومة الروتر بينما **$(\mathbf{R_2/s})$ $(\mathbf{1\text{-}s})$** تمثل مقاومة تعتمد على السرعة والطاقة الناتجة من المحرك

Per-phase output power (assuming that converted power is equal to the output power which means that constant losses are neglected for simplicity of calculations):

$$P_2 = I_2'^2 \frac{R'_2}{S}(1-S) = I_2'^2 \frac{R'_2}{S} - I_2'^2 R'_2$$

Motor output power:

$$P_2 = P_{AG} - \Delta P_2$$

So,

$$P_2 = T\omega_0 - 3I_2'^2 R'_2 = 3I_2'^2 \frac{R'_2}{S} - 3I_2'^2 R'_2 = 3I_2'^2 R'_2 \frac{1-S}{S}$$

Power and torque in an induction motor:

From the equivalent circuit π - type:

Stator current, magnitude of the rotor current, losses in the stator circuit and the power in the air gap can be found as:

$$\dot{I}_1 = \frac{V_{ph}}{Z_{eq}} = V_{ph}\left[\frac{1}{R_m + jX_m} + \frac{1}{(R_1 + \frac{R'_2}{S}) + j(X_1 + X'_2)}\right] \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (1)$$

$$I_2' = \frac{V_{ph}}{\sqrt{(R_1 + \frac{R_2'}{S})^2 + (X_1 + X'_2)^2}} \quad \ldots\ldots(2)$$

$$\Delta P_1 = 3I_1^2 R_1; \Delta P_{core} = 3E_1^2 / R_c \quad \ldots\ldots(3)$$

$$P_{AG} = P_{in} - P_{scl} - \Delta P_{core} = 3I_2'^2 (\frac{R_2'}{S}) \quad \ldots\ldots(4)$$

Losses in the rotor circuit is

$\Delta P_2 = 3I_2'^2 R_2'$ ..... (5)

$P_{ag} - P_2 = \Delta P_2$ ..... (6)

$T\omega_0 - T\omega = T(\omega_0 - \omega) = T\omega_0 S = 3I_2'^2 R_2'$ ..... (7)

From eqs.(2) and (7), the developed motor torque can be obtained as:

$$T = \frac{3I'^2_2 R_2'}{\omega_0 S} = \frac{3V_{ph}^2 R_2'}{\omega_0 S\left[\left(R_1 + \frac{R_2'}{S}\right)^2 + (X_1 + X_2')^2\right]} \quad \ldots\ldots(8)$$

By differentiating the torque equation and taking [dT/ds=0]

Critical slip (s_{cr}) is found to be:

$$Scr = \pm \frac{R_2'}{\sqrt{R_1^2 + (X_1 + X_2')^2}} \quad \ldots\ldots(9)$$

Substituting the value of S_{cr} into equation (8), we obtain equation for maximum torque (critical torque):

$$T_{cr} = T_{max} = \frac{3V_{ph}^2}{2\omega_0(R_1 \pm \sqrt{R_1^2 + (X_1 + X_2')^2}} \quad \ldots.(10)$$, Where the sign (+) corresponding to motoring or braking modes, and the (-) to generative mode.

If eq.(8) Is divided by eq.(10), the torque equation takes the form:

$$T = \frac{2T_{max}(1 + aS_{cr})}{\frac{S}{S_{cr}} + \frac{S_{cr}}{s} + 2aS_{cr}} \quad \ldots\ldots(11)$$

where:

$$a = \frac{R_1}{R_2'}$$

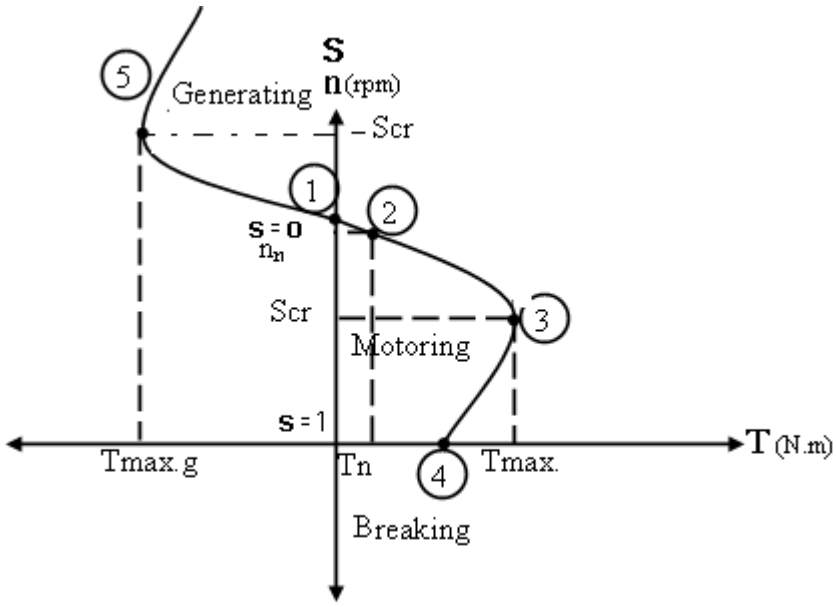
From equation (8), it's important to note that the torque developed by the motor will be proportional to the square of the voltage applied to it.

This means that the motor torque will be sensitive to fluctuation of the supply voltage.

Mechanical Characteristics

The following Figure shows the motor speed torque characteristic.

The points of the particular interest along this curve are (modes):



1- S=0, T=0, n=n_{0°, (motor run at synchronous speed).

2- S=S_n, T=T_n, the point corresponding to rated speed and rated torque.

3- S =S_{cr}, T=T_{max}, (point of maximum torque on motor operation)

4- S=1, (The starting torque point).

$$T_{str} = \frac{2T_{\max}(1+aS_{cr})S_{cr}}{1+S_{cr}(1+2a)} \text{.....(12)}$$

5- S= -S_{cr}, T= -T_{max}, Maximum torque in generative condition.
At slip S>1 , the motor operation in a counter current braking .
It is noted that the absolute value of the Scr would be identical for both motoring and generating condition, but maximum torque are different.

$$T_{max.g} = T_{max.m}\frac{R_1+\sqrt{R_1^2+X_{cr}^2}}{R_1-\sqrt{R_1^2+X_{cr}^2}} \text{.....(13)}$$

Where: $X_{cr} = X_1+X_2'$.
If the stator resistance can be neglected in equation (11), we obtain an equation that becomes more convenient for use in making calculations, namely:

$$T = \frac{2T_{\max}}{\frac{S}{S_{cr}}+\frac{S_{cr}}{S}} \text{.....(14)}$$

$$T_n = \frac{2T_{\max}}{\frac{S_n}{S_{cr}}+\frac{S_{cr}}{S_n}} \Rightarrow \frac{S_n}{S_{cr}}+\frac{S_{cr}}{S_n} = \frac{2T_{\max}}{T_n} = 2\lambda$$

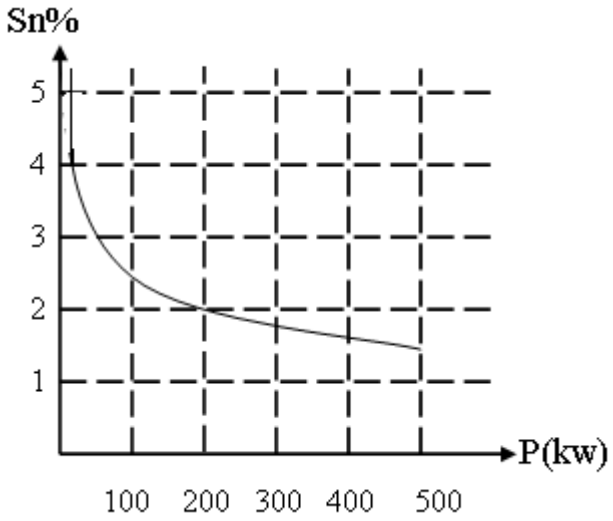
So,

$$\frac{S_n^2 + S_{cr}^2}{S_n S_{cr}} = 2\lambda \Rightarrow S_{cr}^2 - 2\lambda S_n S_{cr} + S_n^2 = 0$$

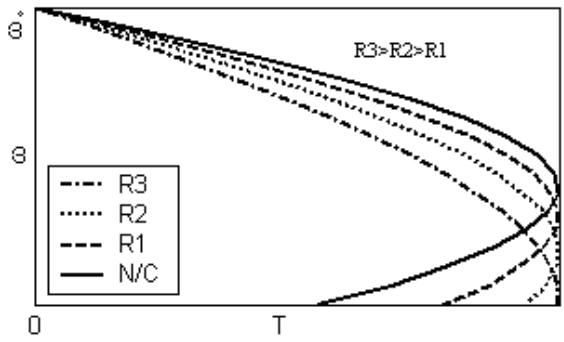
$$S_{cr} = \frac{2\lambda S_n \pm \sqrt{4\lambda^2 S_n^2 - 4S_n^2}}{2} = \lambda S_n \pm S_n\sqrt{\lambda^2 - 1} = S_n(\lambda \pm \sqrt{\lambda^2 - 1})$$

Sign (+) should be used here because (-) is available when $s \rangle s_{cr}$ and this is not used in practice.

S_n- depends on the power of the motor. S_n = 0.005 ÷ 0.15

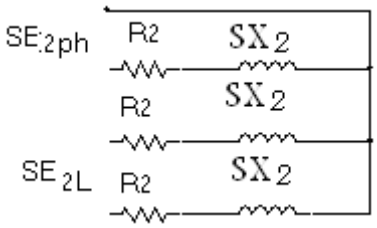


As seen from equation (10). The maximum torque doesn't depend on the rotor resistance R_2. The slip at maximum torque however, is shown by Eq. (9) to increase directly with rise in rotor resistance.
Each increase in rotor resistance will cause the maximum torque point to shift to a position of greater slip as shown in below:



The values of R_2 needed for building the speed- torque characteristics of a slip- ring motor are determined from the expression

$$R_2 = \frac{S_n E_{2L}}{\sqrt{3} * I_{2n}} \quad \text{..... (16)}$$



Maximum to nominal torque ratio (λ)

1. for general- purpose Squirrel- cage motors, λ ≥1.7
2. for general-purpose slip ring motors, λ≥1.8
3. for motors used for cranes λ =2.3 ÷ 3.4

$\frac{T_{str}}{T_n} = 1 \div 1.8$ for squirrel cage motor

$\frac{T_{str}}{T_n} \geq 1.8$ for slip ring motors

$\frac{I_{str}}{I_n} = 5 \div 7$ for squirrel cage motor

Type of motor	(T max/Tn)=λ	Tstr/Tn	Istr/In
Squirrel- cage	≥1.7	1÷ 1.8	5÷7
Slip-ring	≥1.8	≥1.8	variable
Sq. cage motors for cranes	2.3÷3.4	≥1.7	

Motors torque is proportional to the magnetic flux ϕ and current $I`_2$

$T = K\varphi I_2' cos\,\phi_2$…….. (17)

Where K is a coefficient depends on the motor construction, $cos\,\phi_2 = \frac{R_2'}{\sqrt{(R_2')^2+(x_2')^2 S^2}}$

When we raise S, SE_{2L} rises and also I'_2, cos φ_2 lowers, flux φ lowers because of voltage drop at stator winding resistance. All of these factors cause the absence of proportionality between T and I'_2 at starting period. If we change the terminal voltage (stator voltage) , we must keep the rotor losses constant (equal the losses at motor ratings), so , $\Delta P_2 = T_n \omega_0 S_n = T \omega_0 S$. Therefore, $\boldsymbol{T = T_n \frac{S_n}{S}}$

Speed- current (electro-mechanical) characteristics

- $I'_2 = \frac{v_1}{\sqrt{(R_1+\frac{R'_{2\Sigma}}{S})^2+(X_1+X_2')^2}}$……(1)

Where $V_1=V_{ph}$

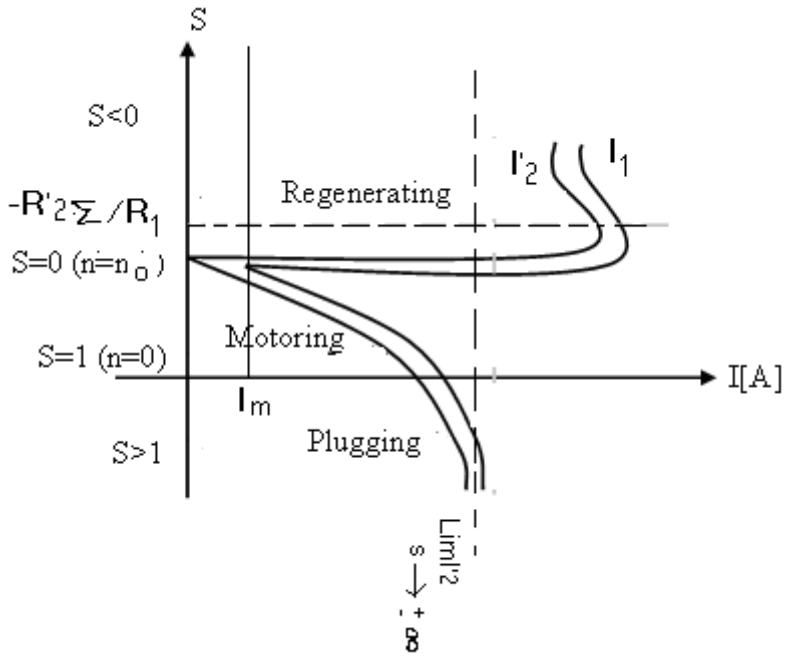
According to the Eq. (1), the following conditions can be found:

1. S=0, $I_2'=0$
2. S=1, $I_2'=I'_{2sc}=\frac{V_1}{\sqrt{(R_1+R'_{2\Sigma})^2+(X_1+X_2')^2}}$
3. S→ ∞ , $I'_2= \lim_{S\to\pm\infty} I'_2 = \frac{V_1}{\sqrt{R_1^2+(X_1+X_2')^2}}$

I_m=constant= (0.15-0.75) I_1; V_1=constant; I_m= I_1 at no-load condition

$I_1=I_m+I'_2$…….. (2)

4. S<0; $I'_{2max} = \frac{V_1}{X_1 + X_2'}$; $R_1 + \frac{R_{2\Sigma}}{S} = 0 \Rightarrow S = -\frac{R'_{2\Sigma}}{R_1}$



Example:-

A 2.8Kw, 400V, 50Hz, 4pole, 1370 RPM, three –phase Squirrel -cage delta- connected motor has the following parameters:

$R_1=R_s=2\ \Omega$, $R_r`=R_2`=5\ \Omega$, $X_1=X_s=5\ \Omega$, $X_r`=X_2`=5\ \Omega$, $X_m=80\ \Omega$.

a.Calculate the λ-ratio and the maximum motor current for this motor

b. Calculate motor speed and current for the terminal voltage of 300V

Solution:

$$\text{Sn} = \frac{\omega_o - \omega_n}{\omega_\circ} = \frac{n_\circ - n}{n_\circ} = \frac{1500-1370}{1500} = 0.0867$$

$$Tn = \frac{3.Vn^2.R_2`}{\omega\circ.Sn[(R1+R2`/Sn)^2 + (X1+X2`)^2]} = \frac{3*400^2*5}{50\pi*0.0867[(2+5/0.0867)^2 + (5+5)^2]} = 48.13N.m$$

$$T_{\max} = \frac{3V_{ph}^2}{2\omega_0(R_1 \pm \sqrt{R_1^2 + (X_1 + X`_2)^2})} = \frac{3*400^2}{2*50\pi[(2+\sqrt{2^2+(5+5)^2}]} = 125N.m, \lambda = \frac{T_{\max}}{T_n} = \frac{125}{48.13} = 2.6$$

$$I'_{2\max} = \frac{V_{ph}}{\sqrt{(R_1 + R'_2)^2 + (X_1 + X'_2)^2}} = \frac{400}{\sqrt{(2+5)^2 + (5+5)^2}} = 32.8A$$

$$I_{mag} = \frac{V_{ph}}{X_m} = \frac{400}{80} = 5A, then I_{s\max} = I_{mag} + I'_2 = 5 + 42.8 = 47.8A$$

$$Tn.Sn = Tper.S$$

$$48.13*0.0867 = \frac{3(300)^2*5}{50\pi[(2+5/S)^2 + (10)^2]} = \frac{27*10^4*5^2}{10\pi(1048^2 + 20S + 25)}$$

$$195.5S^2 - 20S - 25 = 0 \Rightarrow S = 0.1183, \text{n} = \text{n}\circ(1-\text{S}) = 1500(1-0.1183) = 1322\text{RPM}$$

$$\text{Is} = \text{I1} = \text{Ir`} + \text{Im} = \frac{\text{VT}}{(\text{R1}+\text{R2`/S}) + \text{j}(\text{X1}+\text{X2`})} + \frac{VT}{jXm12} = \frac{300}{(2+5/0.1183) + j10} + \frac{300}{j80}$$

$$= 300*[\frac{1}{44.26 + j10} + \frac{1}{j80}] = 6.45 - j5.2$$